Program Design Quiz 5

(Please use C language to solve the problem)

題目敘述：

給定 N 組整數陣列。第 i 組陣列先輸入一個整數 m_i，代表該組元素個數，接著輸入 m_i 個整數。請將所有組別的元素合併後，依下列規則輸出：

1. 中位數：
   - 合併後長度 M 為奇數：輸出序列排序後的正中央元素（整數格式）。
   - M 為偶數：輸出排序後中間兩個元素的平均值（若值含小數，輸出到小數點後一位；若為整數，則輸出整數格式）。
2. 眾數：
   - 在合併後序列中，出現次數最多的值。若有多個眾數，請以遞增順序輸出、以單一空白分隔。

輸入說明：

- 第 1 行輸入一個整數 N（組數）。
- 接下來共 N 組輸入：每組先輸入一個整數 m_i，接著輸入 m_i 個整數。
- 所有整數皆在 -2^{31} 與 $2^{31}-1$ 之間；且保證合併後的總元素數 $M=\Sigma m_i \geq 1$。

輸出說明：

第 1 行輸出中位數：

- M 為奇數：輸出整數（例如 3）。
- M 為偶數：依據中位數數值輸出到小數點後一位或整數（例如 2 或 3.5）。

第 2 行輸出眾數：

- 若有多個眾數，以遞增順序、以單一空白分隔輸出（例如 1 2 4）。
- 最後結束需要換行

Examples

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Input 1:

2

3 1 3 5

2 3 9

Output 1:

3

3

Input 2:
3
2 4 4
3 1 2 2
1 1

Output 2:
2
1 2 4

Input 3:
3
3 4 4 3
4 1 2 2 4
1 1

Output 3:
2.5
4

Program Design Quiz 5

(Please use C language to solve the problem)

Problem Description：

Given N arrays of integers. For the i array, you first input an integer mi denoting the number of elements in that array, followed by mi integers. After merging all arrays, output the following:

1. Median
   - If the merged length M is odd: output the middle element of the sorted sequence (as an integer).
   - If M is even: output the average of the two middle elements after sorting. If the value is fractional, print it to one decimal place; if it is an integer, print it in integer format.
2. Mode(s)
   - The value(s) that appear most frequently in the merged sequence. If there are multiple modes, output them in increasing order, separated by a single space.

Input：

- The first line contains an integer N (the number of arrays).
- Then follow N groups of inputs: for each group, first an integer mi, then mi integers.
- All integers are within the range $[-2^{31}, 2^{31}-1]$.
- It is guaranteed that the total number of elements after merging is $M=\sum mi \geq 1$.

Output：

Line 1 (Median):

- If M is odd, print the median as an integer (e.g., 3).
- If M is even, print either an integer or a value with one digit after the decimal point, depending on the median's value (e.g., 2 or 3.5).

Line 2 (Mode):

- If there are multiple modes, print them in increasing order, separated by a single space (e.g., 1 2 4).
- The output must end with a newline.

Examples

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Input 1:

2

3 1 3 5
2 3 9

Output 1:
3
3

Input 2:
3
2 4 4
3 1 2 2
1 1

Output 2:
2
1 2 4

Input 3:
3
3 4 4 3
4 1 2 2 4
1 1

Output 3:
2.5
4